higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 3

(6021033)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat and in proportion
5. Clearly cross out ALL the work that you do not want to be marked.
6. Start each question on a NEW page.
7. Write neatly and legibly.

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QUESTION 1

1.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–F) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

	COLUMN A		COLUMN B
1.1.1	Negative camber	A	outward tilt of the top of the wheel
1.1.2	Steering axis inclination	B	a pre-check before checking wheel alignment readings
1.1.3	Caster	C	forward or backward tilt of the top of the strut
1.1.4	Drive height	D	inward tilt of the top of the wheel
1.1.5	Over-steer	E	a non-adjustable wheel alignment angle
		F	the driver must turn the steering wheel back to rectify it

(5 × 1) (5)

1.2 List FIVE steps to follow when dismantling a MacPherson strut assembly to replace the shock absorber after it has been removed from the car. (5)

1.3 Identify THREE faults that can occur on the front suspension parts due to normal wear. (3)

1.4 You inspected a wheel that a customer wanted you to balance and decided to inform the customer that the wheel could not be balanced.

Give FOUR reasons why it could not be balanced. (4)

1.5 Name the TWO types of balancing that must be done to ensure that the complete wheel is balanced. (2)

1.6 Which type of safety guard is used by a wheel balancing machine? (1)

1.7 List THREE parts of the wheel balancing machine that you would use to complete the balancing task. (3)

[23]

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QUESTION 2

2.1 State the FIVE basic steps to follow when dismantling a starter motor after it has been removed from the car. (5)

2.2 Name THREE tests that should be done on a starter motor before installing it on the vehicle. (3)

2.3 Study the picture in FIGURE 1 below and answer the questions.

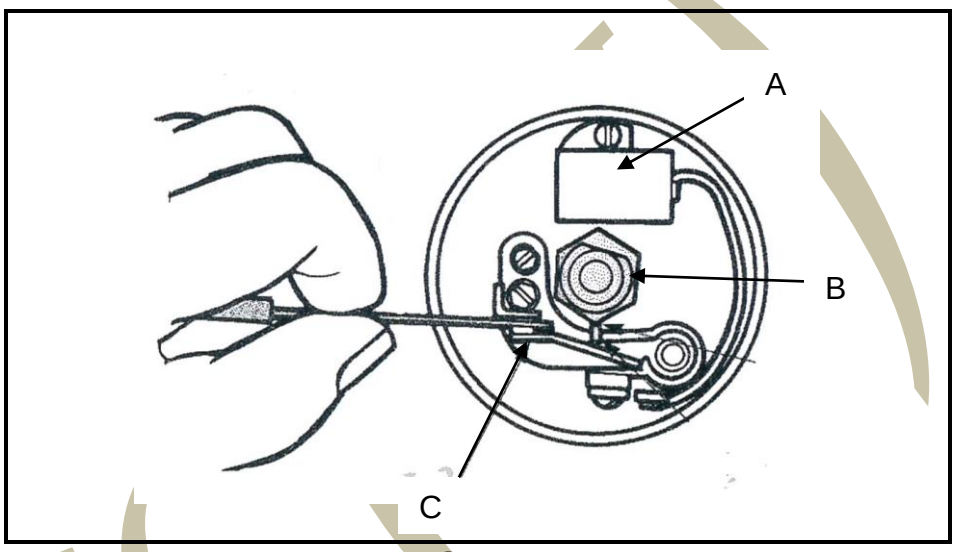


FIGURE 1

2.3.1 Identify the task that is carried out. (1)

2.3.2 Which angle is affected by the task in QUESTION 2.3.1? (1)

2.3.3 State TWO reasons why the angle mentioned in QUESTION 2.3.2 could be inconsistent. (2)

2.3.4 Name the parts that are labelled A B and C. Write only the answer next to the letter (A–C) in your ANSWER BOOK). (3)

2.3.5 Name the tool that is used for the task in FIGURE 1. (1)

2.4 Name the TWO parts of the alternator that is responsible for producing the output voltage before it is rectified. (2)

2.5 Which parts of the alternator are responsible for the following tasks:

2.5.1 To change the output voltage from AC to DC

2.5.2 To control the amount of output voltage to keep the battery charged

(2 × 1) (2)

2.6 List any FOUR parts of a basic charging system of a motor vehicle. (4)

[24]

QUESTION 3

3.1 Briefly explain the purpose of the following additives used in petrol:

3.1.1 Metal deactivators

3.1.2 Anti-icing agents

3.1.3 Detergents

(3 × 1) (3)

3.2 What is meant by a s*toichiometric* fuel mixture? (1 × 2) (2)

3.3 A vehicle has a fuel starvation problem.

List THREE possible reasons why it would not get enough fuel in the float bowl of the carburettor. (3)

3.4 Briefly explain why diesel engines use glow plugs. (2)

3.5 Explain how the difference in the diameters of brake wheel cylinders influences the brake force. (3)

3.6 Identify the parts labelled A, B, C, D and E in FIGURE 2 below. Write only the answer next to the letter (A–E) in your ANSWER BOOK).

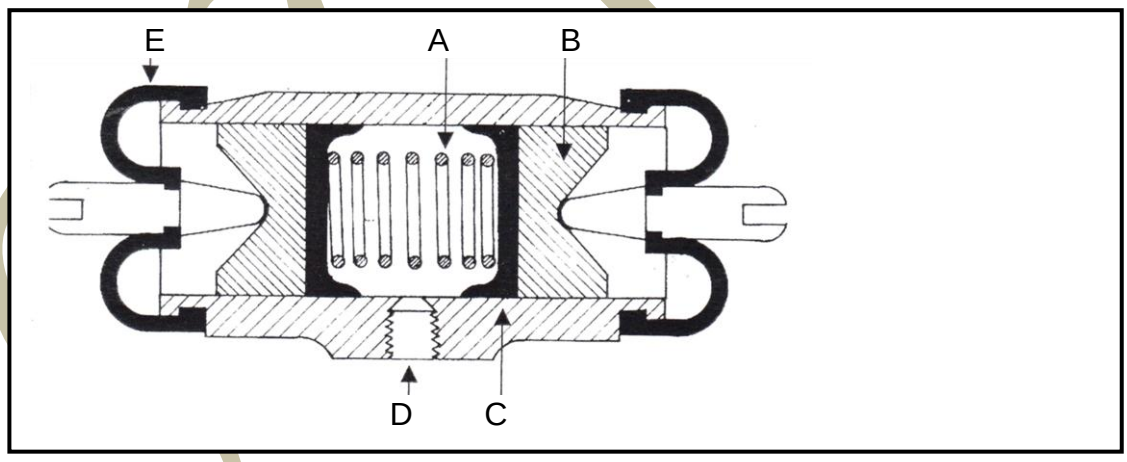


FIGURE 2 (5)

3.7 Make a simple sketch of each of the following to differentiate between their tooth patterns. Label each sketch with the appropriate name.

3.7.1 Helical gear

3.7.2 Spur gear

3.7.3 Double helical gear

(3 × 2) (6)

3.8 Name the TWO methods used to bleed a brake system. (2)

3.9 Name TWO specifications that you would require to assemble a final drive. (2)

[28]

QUESTION 4

4.1 Determine THREE possible conditions of the engine if the wet leakage test result is much higher than the dry test result. (3)

4.2 Explain why it is important to clean the engine parts before sending them to the engineering workshop. (2)

4.3 Study FIGURE 3 below and answer the questions.

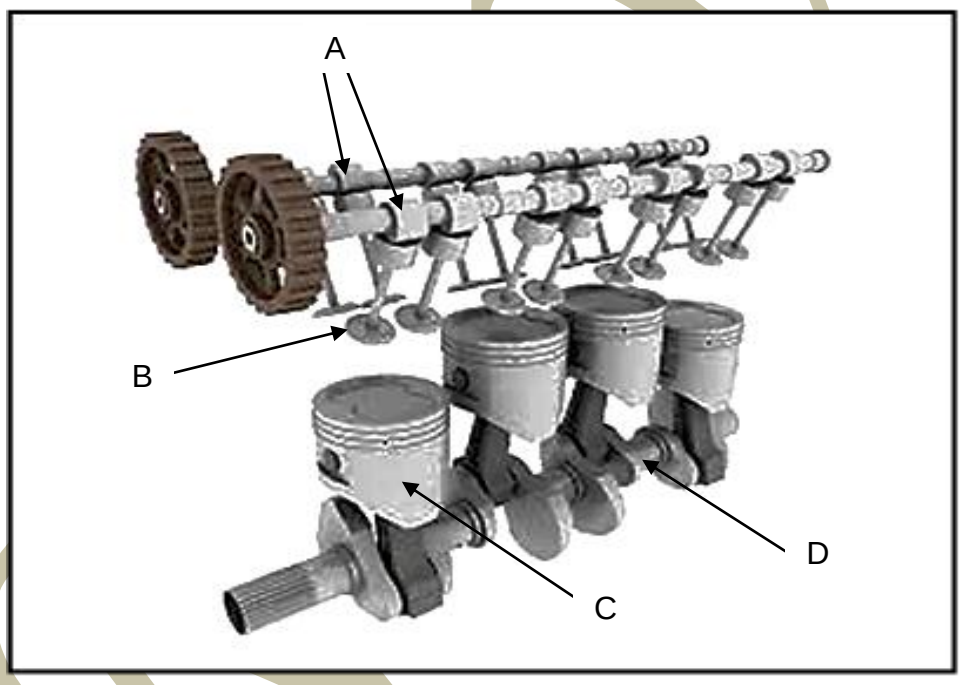


FIGURE 3

Explain the type of repair work necessary under the following conditions:

4.3.1 The lobe of part A is badly worn.

4.3.2 Part B causes a slight loss of compression.

4.3.3 Part C is worn and damaged the cylinder.

4.3.4 Part D is worn at the main journal.

(4 × 2) (8)

4.4 Name TWO steps to follow when preparing an engine for a compression test. (2)

4.5 Describe any THREE methods that can be used to clean engine components for inspection. (3)

4.6 Name FOUR pieces of workshop equipment that you would require to safely remove and store an engine. (4)

4.7 List THREE advantages of using a workshop manual when preparing to remove an engine. (3)

[25]

TOTAL: 100

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